CARDS: A Novel Detailed Macro Placement Framework for Minimizing Wirelength

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Abstract—**In contemporary IC design, the use of reusable intellectual property (IP) and embedded memory is pivotal for expediting design cycles and enhancing performance. A prominent trend in efficient IC design is the adoption of mixed-size circuits, which incorporate with large macros and millions of standard cells. Among the various design stages, macro placement emerges as a critical factor. Strategically organizing macros during this phase can greatly facilitate standard-cell placement and routing, thereby significantly improving circuit permanence. This paper introduces CARDS, a novel framework tailored for detailed macro placement. CARDS enhances an initial macro placement through a series of flipping and moving operations. The framework comprises five steps: Data processing, Recomputed center finding, Adjustment, Sixteen-type flipping, and Center-toward flipping. Our experimental findings showcase the capability of CARDS to achieve a noteworthy reduction of up to 16.54% in half-perimeter wirelength (HPWL) post-detailed routing compared to the initial macro placement.**

Keywords: Detailed macroplacement, Wirelength minimization, Macro flipping, Macro movement, Binary search tree

I. INTRODUCTION

Due to the increasing reliance on IP and embedded memories, advanced mixed-size circuit design, which involves numerous elements including thousands of macros and millions of standard cells, is becoming increasingly important nowadays. The mixed-size circuit design procedure involves three stages: mixed-size prototyping, macro placement, and standard cell placement. Prototyping determines initial positions for macros and standard cells, considering wirelength and routability. Macro placement further optimizes the location and orientation of each macro considering wirelength, routability, displacement, and free spaces for standard cells. Finally, standard cells are placed afterwards.

Among the three stages, macro placement is the most crucial since it significantly impacts the final standard-cell placement, as well as total wirelength and routability. Therefore, previous researchers have proposed various strategies for macro placement, including MP-Tree-based [1], MDP-tree-based [2], and damp-waved-based [3] placement frameworks for reducing total wirelength. However, most of the previous works focused on macro legalization to generate non-overlapped macro placement results. We find that an efficient detailed macro placement algorithm, which can further minimize total wirelength by flipping and adjusting a legalized macro placement, is in demand. This detailed macro placement algorithm should deal with the pin accessing problem by finding appropriately flipping macros and further minimize total wirelength by slightly moving macros.

In this paper, we introduce a novel detailed macro placement framework, CARDS, aimed at further minimizing total wirelength by strategically flipping and relocating macros derived from a given mixed-size prototyping result. Specifically, we employ a strategy of moving macros based on available vacant areas to create free space adjacent to macros for standard cells, while also employing a cost function reflecting the Half Perimeter Wire Length (HPWL) to guide macro flipping decisions. Experimental results demonstrate that our proposed framework can achieve a reduction in HPWL of up to 16.54% compared to the initial macro placement.

II. PROBLEM STATEMENT

In this section, the assumptions, constraints, and optimization goals of the detailed macro placement problem are described. First, each macro has four possible flipping directions: N (no flip), FN (Y-axis flip), S (flip along both axes), and FS (X-axis flip). Second, a maximum displacement constraint is set to maintain prototyping results. Third, all macros must be placed within chip boundaries without overlapping.

Specifically, the detailed macro placement problem can be stated as follows: Given an initial macro and standard cell placement obtained from mixed-size prototyping stage, we have to decide the orientation and location of each macro so that the constraints mentioned above are all satisfied and the total wirelength in HPWL after detailed route should be minimized.

III. DETAILED MACRO PLACEMENT FRAMEWORK

This section details the proposed CARDS framework, which consists of five steps: **D**ata processing, **R**ecomputed center finding, **A**djustment, **S**ixteen-type flipping, and **C**enter-toward flipping. The overview of our framework is shown in Fig. 1. Generally, the proposed method firstly computes the centers of the continuous vacant areas, then moves macros toward the centers based on the distance between original location of macros and the centers. After that, a 16-type flipping algorithm is applied based on the cost function, following by a center-toward flipping method. Finally, the detail macro placement is generated for the following steps of physical design. We detail each step of the proposed framework in the following sub-sections.

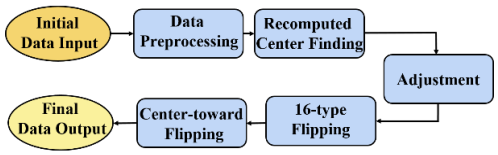

Figure 1. The overview of CARDS framework

A. Data preprocessing

In data preprocessing stage, we read the design information from Verilog files, LEF files, and DEF files after the mixed-size

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prototyping stage. A well-designed map-based data structure with C++ STL are used for efficient sorting and searching.

B. Recomputed center finding

One of the major ideas in the proposed framework is to place standard cells adjacently to the macros which have wires in between. Therefore, sufficient free spaces should be reserve for standard cells during macro placement stage. In our framework, we firstly identify the centers of the continuous areas that are not occupied by macros from initial placement result, and then calculate the centers of these areas. Specifically, the entire design is divided into16 x16 squares, and the center coordinates of each continuous vacant area (as shown in red and blue in Figure 2) can be obtained for later usage.

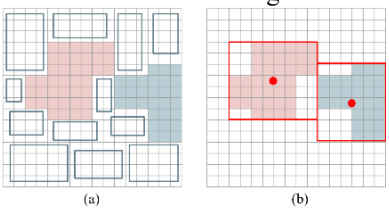

Figure 2. An exmaple of center finding (a) Vacant area (b) Center of continuous vacant area

C. Adjust macros by moving

In this step, the goal is to move macros based on the center points we find in the previous step. Specifically, we find the nearest center point of each macro, and the macro will be moved toward the center point. The moving distance will be decided by the relative distance between all macros that moving toward to the center point. We use the following equations to calculate the moving distance

$$\text{Manhattan} = D_{const.} * \frac{D_{max} - D}{D_{max} - D_{min}} \quad (1)$$

where *Manhattan* represents the moving distance for the macro, $D_{const.}$ denotes the displacement constraint, D_{max} is the distance between the location of farthest macro that will move toward to the center point and the center point, $D_{min\ x}$ is the distance between the location of nearest macro that will move toward to the center point and the center point, and D represents the distance between the location of the target macro and the center point. Based on the equation, the macros closer to the center point will have large move steps, creating free spaces nearby other macros for standard cells.

We build two linked-lists to store the x and y coordinates of each macro, based on their bottom-left and upper-right coordinates. The movement of macros mentioned above can be easily performed on the linked-lists.

D. 16-type flipping

To reduce total wirelength, a binary search tree is optimized from the wire with the highest HPWL. Flipping changes pin locations on macros, reducing wirelength between macros and standard cells. This stage focuses on the macros on the boundary. Each macro has four directions to flip, so two macro pairs on the boundary result in 16 combinations. Using the cost function (2), which considers changes in single HPWL and other wires' HPWL, the direction combination with the smallest cost value is selected. This process avoids local optimal solutions and efficiently reduces wirelength.

$$\text{Cost} = 2 * \Delta(\text{single HPWL}) + \Delta(else\ HPWL) \quad (2)$$

E. Center-toward flipping

Finally, the macros that have not been flipped in previous steps are flipped toward the center in the direction with the highest number of pins.

IV. Experiment Result

To assess the effectiveness of the CARDS framework, NTUplace3 [4] was employed. NTUplace3 is an analytical placer tailored for large-scale mixed-size designs with preplaced blocks and density constraints, making it well-suited for the proposed approach. Half-perimeter wirelength (HPWL) serves as the standard metric for evaluating the total wirelength reduction achieved by the proposed methodology.

Table. I compare the results of three methods across 6 cases with different numbers of movable macros [5]: using only 16-type flipping, a combination of 16-type and center-toward flipping, and our CARDS framework. CARDS consistently achieve a positive reduction percentage, up to 16.54% in case E.

V. Conclusions

This work concentrates on minimizing the total wirelength in the second stage of the three-stage mixed-size placement procedure in IC design. The proposed CARDS framework redefines the center of each macro and adjusts all movable macros towards this center. By considering all 16 types of flipping and employing a cost value to select the optimal type, CARDS achieves notable reductions in wirelength. The findings demonstrate that the CARDS framework can effectively reduce the total wirelength by up to 16.54%.

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Table I. Benchmark Information and HPWL comparisons

Case	#Macro	Original	16-type flipping		16-type + Center-toward flipping		CARDS	
		HWPL	*HPWL*	*reduction*	*HPWL*	*reduction*	*HPWL*	*reduction*
A	175	219,156,226,426	221,247,599,135	-0.95%	212,585,748,733	3.00%	211,040,872,330	3.70%
B	175	210,833,009,088	213,578,609,193	-1.30%	213,876,519,275	-1.44%	207,127,074,347	1.76%
C	84	333,452,406,908	336,829,151,308	-1.01%	343,405,036,063	-2.98%	321,851,648,134	3.48%
D	84	324,831,995,015	323,273,688,146	0.48%	336,861,916,693	-3.70%	317,017,397,012	2.41%
E	84	409,405,420,812	338,380,340,318	17.35%	340,792,184,686	16.76%	341,661,582,061	16.54%

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